



1. GPR Survey

7 Groups × 3 Heights

21 Radargrams



2. GPR Pre-processing

Time-zero correction

Trace alignment

Denoise

Mean-trace removal

T-Power gain ($n = 1$)



3. W1 & W3 Attribute Extraction

W1 (8–18 ns)

W3 (23–33 ns)



4. Selection of Key Attributes

RMS

Energy

SDR

Adjacent-trace correlation



5. Geotechnical Validation & Depth Conversion

Recent BH

Previous BH

CPT-derived v_s

$V = 0.14$ m/s